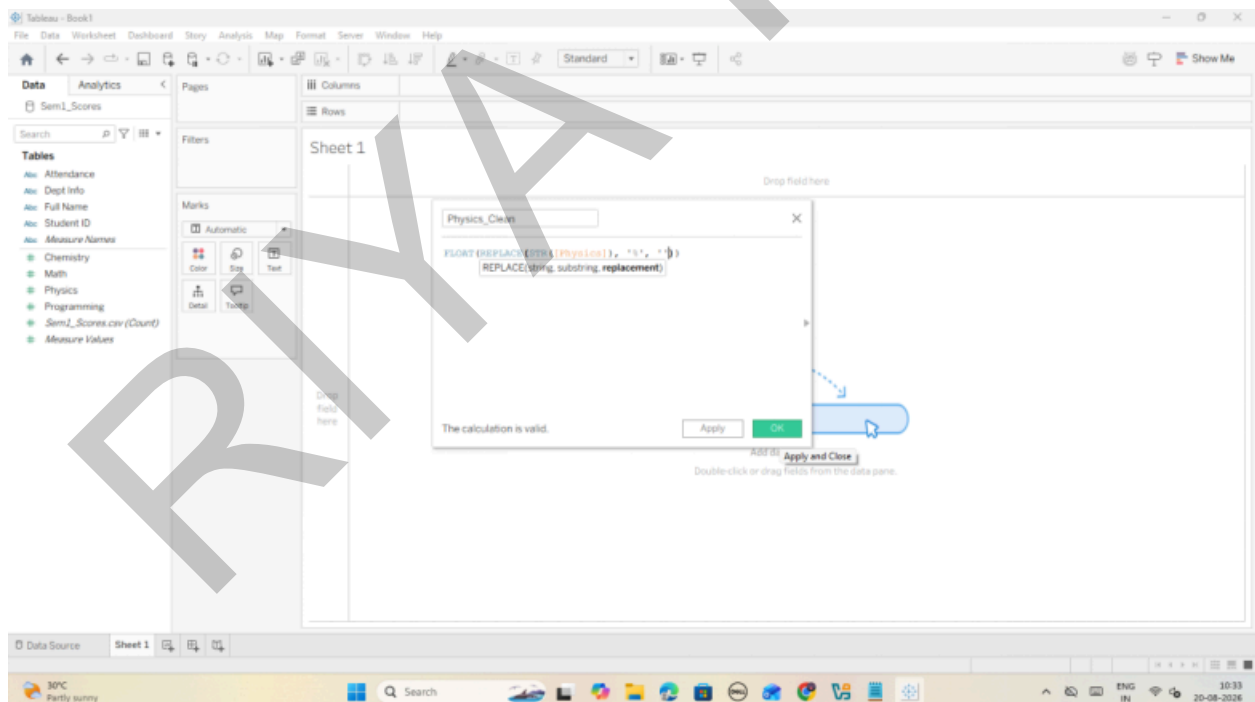
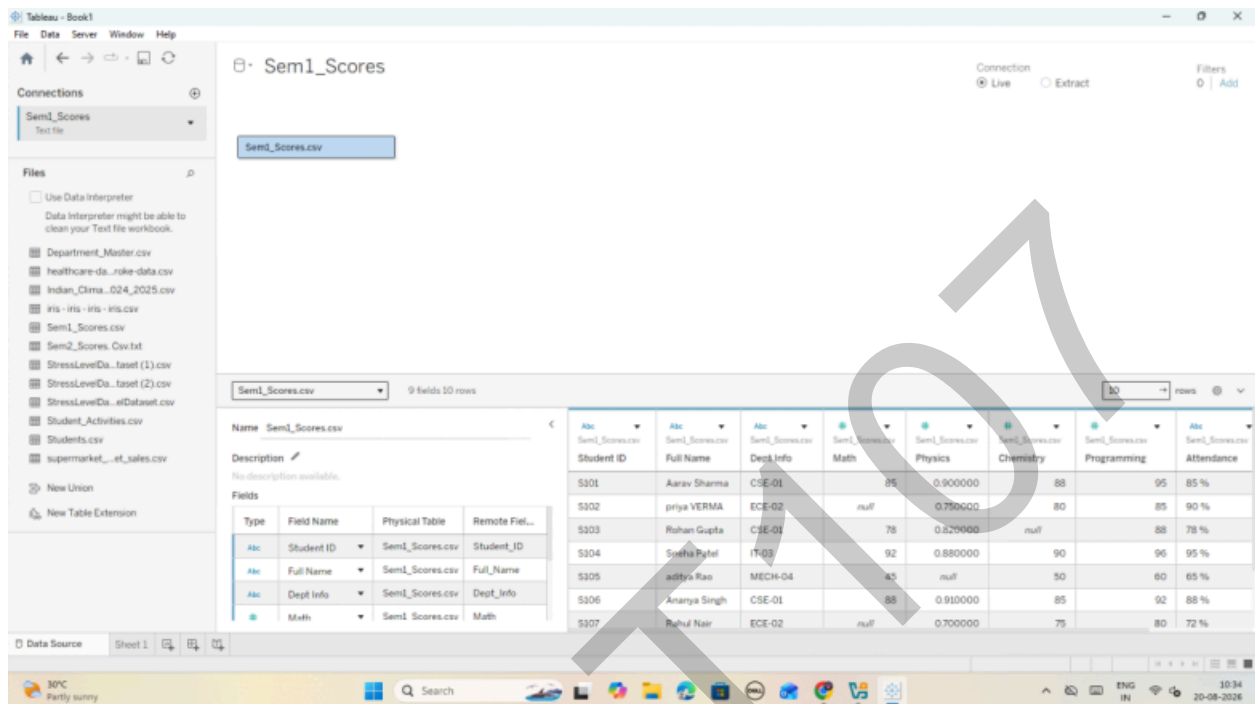


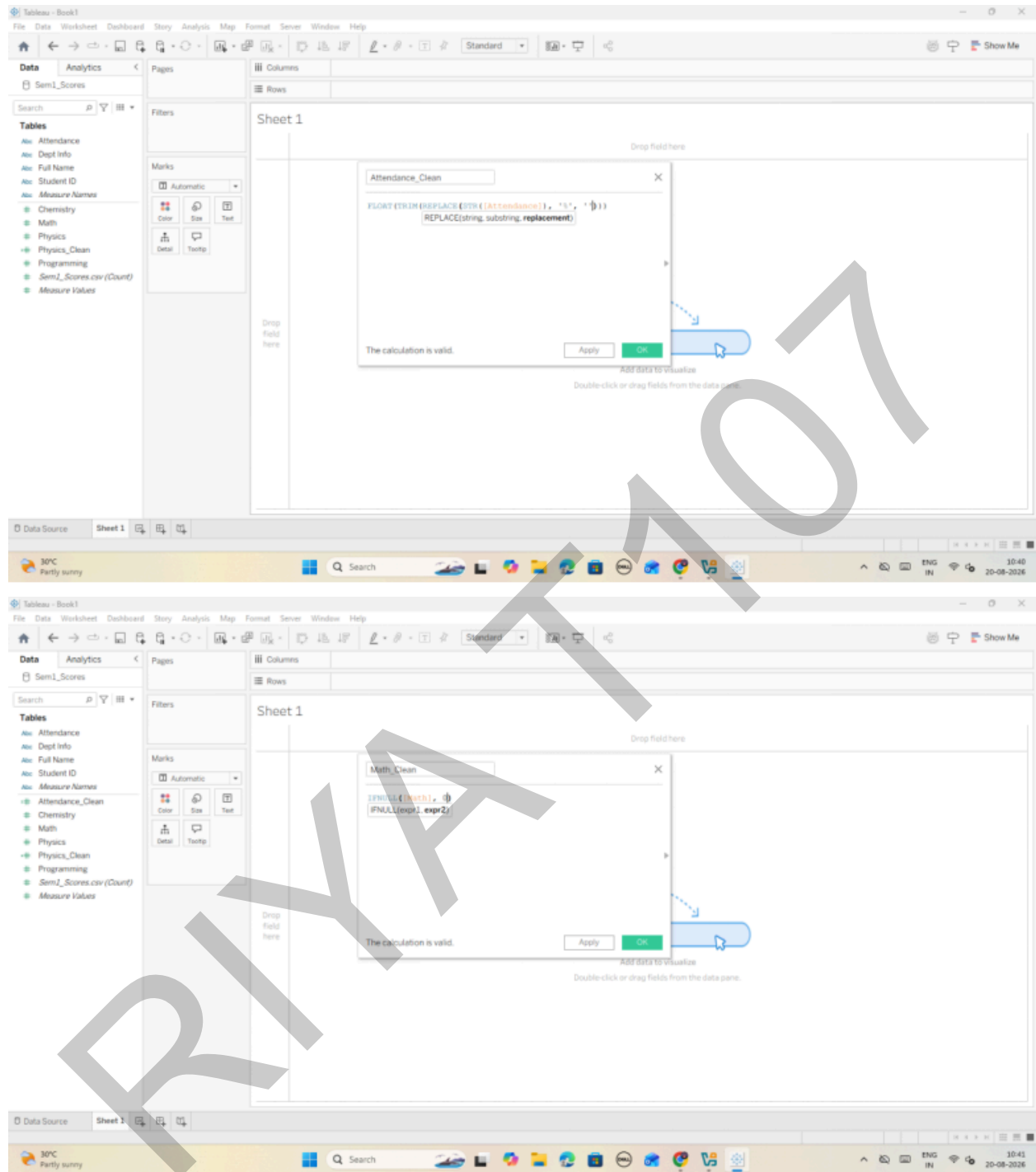
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AIM: Cleaning and Structuring Data for Analysis

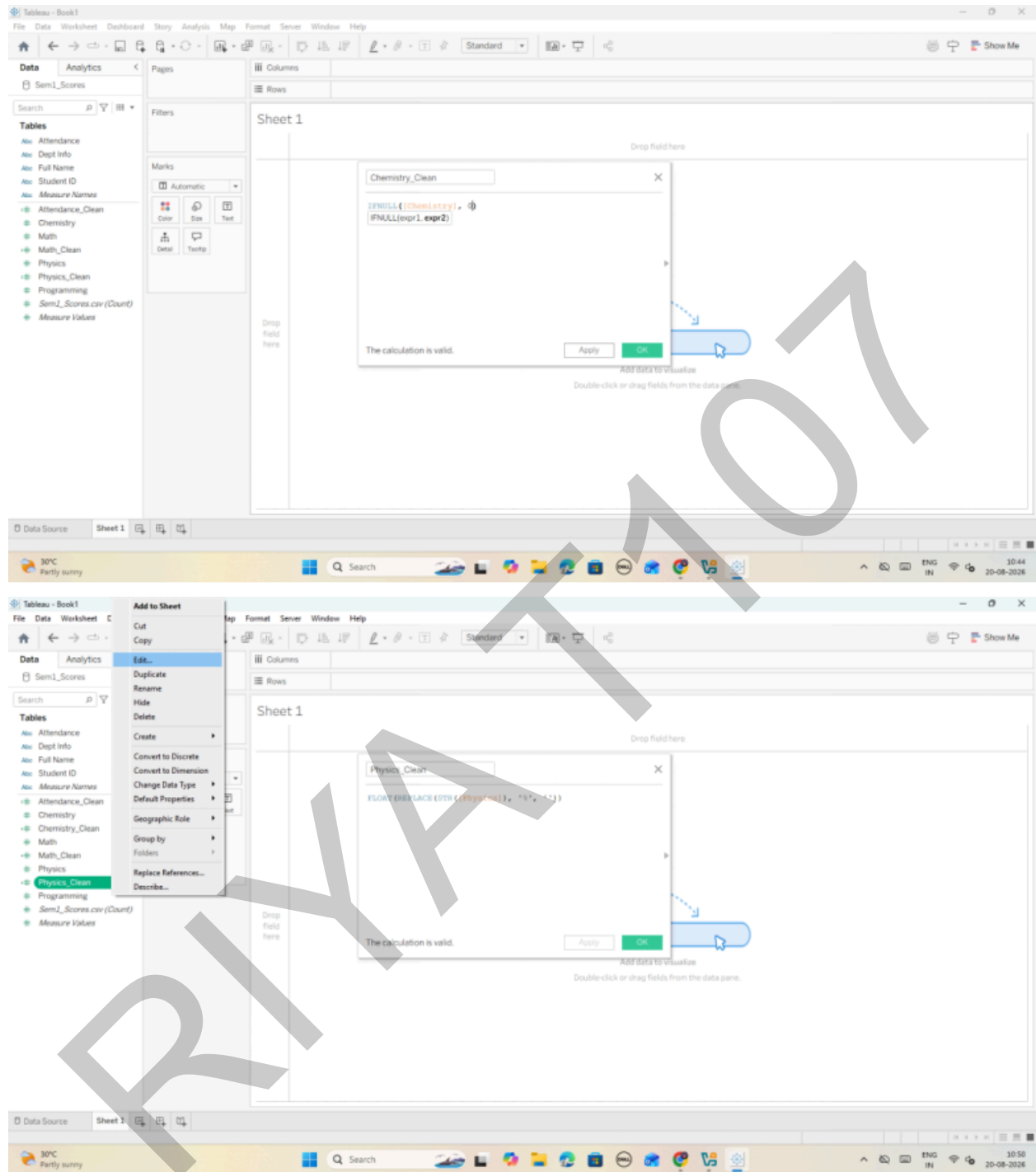
- Identify and handle missing or inconsistent data
- Convert wide data into tall format



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The top screenshot shows the Tableau Desktop interface with a calculation editor open for 'Physics_Clean'. The formula entered is: `IFNULL (FLOAT (REPLACE (STR ([PhysLoc]), ' ', '')), 0)`. The bottom screenshot shows the 'Sem1_Scores' data source with a table view of student scores. The table has columns for Student ID, Full Name, Dept Info, Attendance, Post Field Names, Post Field Values, and various subject scores (Physics_Clean, Attendance_Clean, Math_Clean, Chem).

Student ID	Full Name	Dept Info	Attendance	Post Field Names	Post Field Values	Physics_Clean	Attendance_Clean	Math_Clean	Chem
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	88.0000	85.0000	85.0000	
S001	Aarav Sharma	CSE-01	85 %	Math	85.0000	85.0000	85.0000	85.0000	
S001	Aarav Sharma	CSE-01	85 %	Physics	0.0000	85.0000	85.0000	85.0000	
S001	Aarav Sharma	CSE-01	85 %	Programming	95.0000	85.0000	85.0000	85.0000	
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	80.0000	90.0000	90.0000	
S002	priya VERMA	ECE-02	90 %	Math	80.0000	80.0000	90.0000	90.0000	
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	80.0000	90.0000	90.0000	
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	85.0000	90.0000	90.0000	
S003	Rohan Gupta	CSE-01	78 %	Chemistry	78.0000	78.0000	78.0000	78.0000	

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The image displays two screenshots of the Tableau Desktop interface, specifically the 'Sem1_Scores' data source. The top screenshot shows the 'Fields' shelf with 'Score' as a dimension. The bottom screenshot shows the 'Fields' shelf with 'Score' as a measure, and a context menu is open over the 'Score' field.

Tableau Desktop - Sem1_Scores

Connections: Sem1_Scores (Text file)

Files: Department_Master.csv, healthcare-da_roke-data.csv, Indian_Clima_024_2025.csv, iris - iris - iris.csv, Sem1_Scores.csv, Sem2_Scores.csv.txt, StressLevelDa_taset (1).csv, StressLevelDa_taset (2).csv, StressLevelDa_etDataset.csv, Student_Activities.csv, Students.csv, supermarket_et_sales.csv

Fields:

Type	Field Name	Physical Table	Remote Field
Dimension	Score	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Student ID	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Full Name	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Dept Info	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Attendance	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Subject	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Physics_Clean	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Attendance_Clean	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Math_Clean	Sem1_Scores.csv	Sem1_Scores.csv
Dimension	Chem	Sem1_Scores.csv	Sem1_Scores.csv

Data Source: Sem1_Scores (Text file)

Table: Sem1_Scores (Text file)

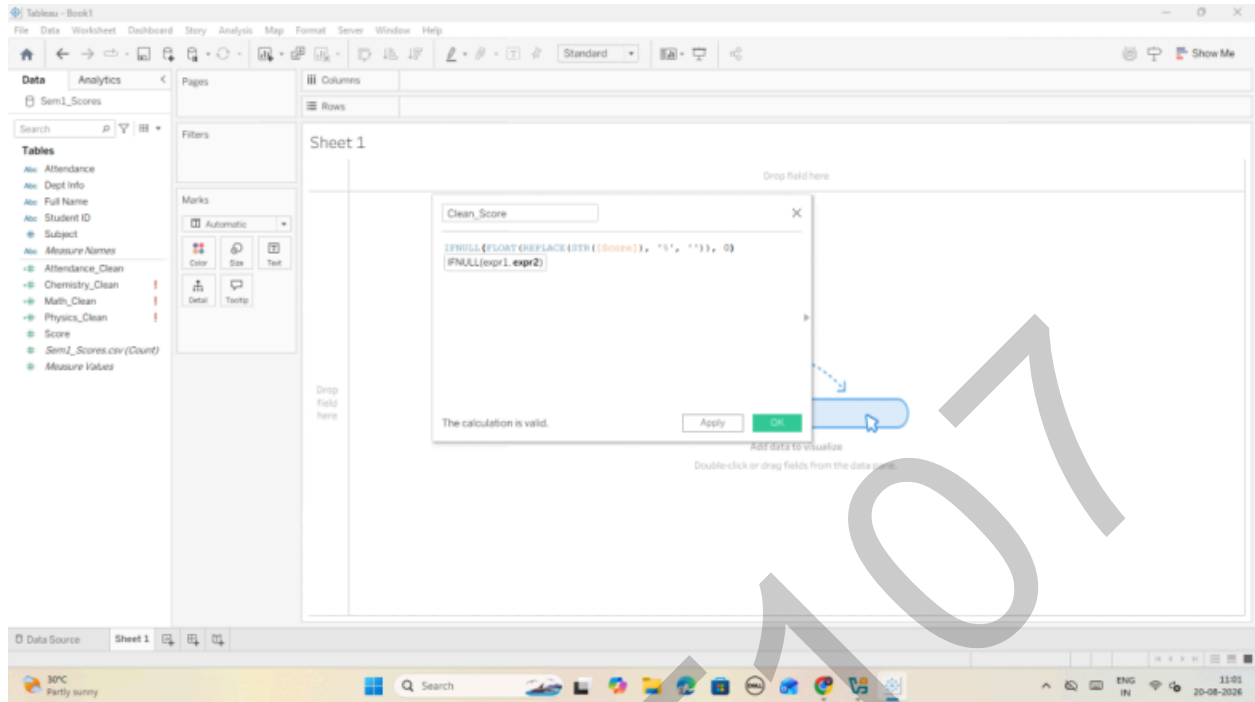
Columns: Student ID, Full Name, Dept Info, Attendance, Subject, Score, Physics_Clean, Attendance_Clean, Math_Clean, Chem

Rows: 40 rows

Score Field Context Menu:

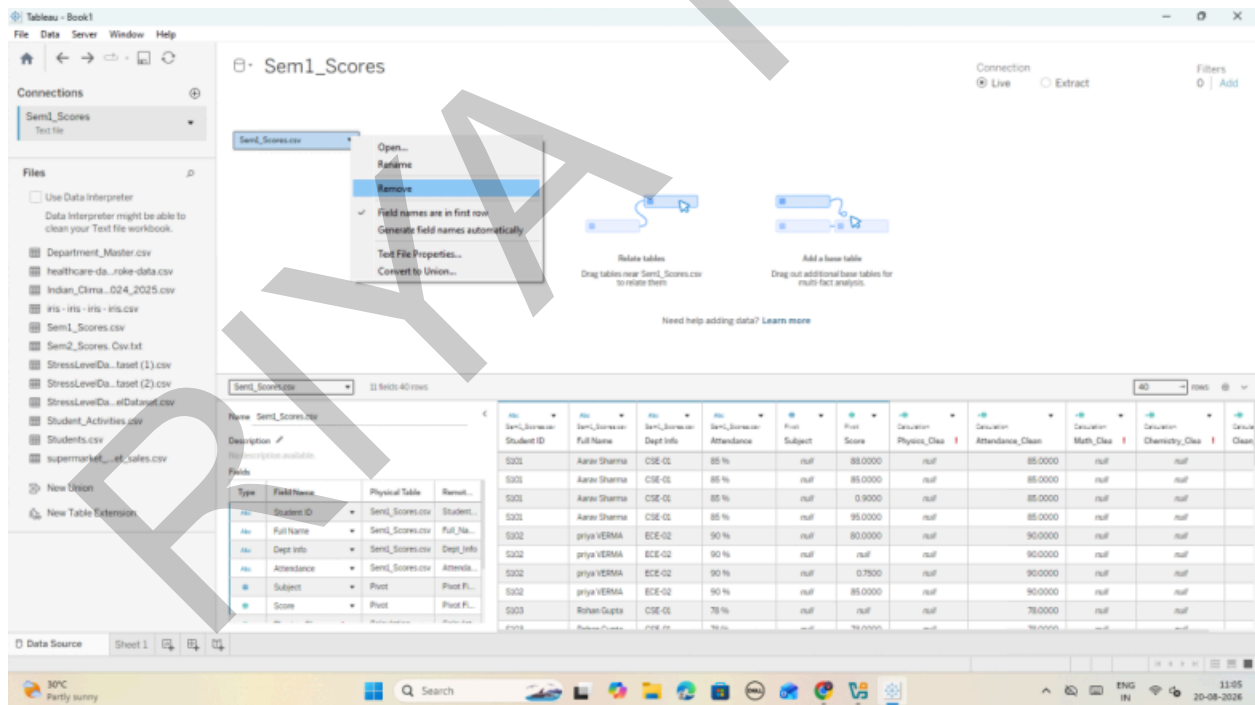
- Number (Decimal)
- Number (Integer)
- Date & Time
- String
- Geographic Role
- Image Role
- Score
- Geographic Role
- Image Role
- Score

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c) Perform union of multiple files

d) Execute cross-database joins



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The screenshot displays the Tableau Desktop interface. On the left, the 'Connections' pane shows 'Sem1_Scores' as a 'Text file'. The 'Files' pane lists various CSV files, including 'Sem1_Scores.csv'. A 'Union' dialog box is open, showing 'Sem1_Scores' as the connection. The 'Specific (manual)' tab is selected, and 'Sem1_Scores.csv' is listed under 'Tables in union: 2'. The 'Apply' button is highlighted. Below the dialog, the 'Data Source' pane shows 'Sem1_Scores' as a 'Text file'. The main view shows a table with columns: 'Student ID', 'Full Name', 'Dept Info', 'Math', 'Physics', 'Chemistry', 'Programming', 'Attendance', and 'PHYSICS_CLEAN'. The table contains 8 rows of data.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	PHYSICS_CLEAN
S001	Aarav Sharma	CSE-01	85	0.900000	88	95	85%	0.900000
S002	priya VERMA	ECE-02	null	0.750000	80	85	90%	0.750000
S003	Rohan Gupta	CSE-01	78	0.820000	null	88	78%	0.820000
S004	Sneha Patel	IT-03	92	0.880000	90	96	95%	0.880000
S005	aditya Rao	MECH-04	45	null	50	60	65%	0.000000
S006	Ananya Singh	CSE-01	88	0.980000	85	92	88%	0.980000
S007	Rahul Nair	ECE-02	null	0.700000	75	80	72%	0.700000
S008	Kavya Reddy	IT-03	95	0.940000	92	98	98%	0.940000

The image displays two screenshots of the Tableau Desktop interface, illustrating the process of creating a multi-fact analysis.

Top Screenshot: The interface shows the 'Sem1_Scores' worksheet. The 'Connections' pane on the left lists 'Sem1_Scores' as a Text file. The 'Files' pane shows a list of data sources, including 'Sem1_Scores.csv', 'Sem2_Scores.csv', and 'Sem3_Scores.csv'. The 'Data Source' pane shows 'Sem1_Scores.csv' with 12 fields and 10 rows. The 'Fields' pane shows the 'Sem1_Scores.csv' table with fields: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, and Programming. The 'Columns' shelf contains 'Sem1_Scores.csv' (Student ID), 'Sem1_Scores.csv' (Full Name), 'Sem1_Scores.csv' (Dept Info), 'Sem1_Scores.csv' (Math), 'Sem1_Scores.csv' (Physics), 'Sem1_Scores.csv' (Chemistry), 'Sem1_Scores.csv' (Programming), and 'Calculation' (PHYSICS_CLEAN). The 'Rows' shelf contains 'Calculation' (PHYSICS_CLEAN).

Bottom Screenshot: The interface shows the 'Sem1_Scores' worksheet. The 'Connections' pane on the left lists 'Sem1_Scores' as a Text file. The 'Files' pane shows a list of data sources, including 'Sem1_Scores.csv', 'Sem2_Scores.csv', and 'Sem3_Scores.csv'. The 'Data Source' pane shows 'Sem1_Scores.csv' with 12 fields and 10 rows. The 'Fields' pane shows the 'Sem1_Scores.csv' table with fields: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, and Programming. The 'Columns' shelf contains 'Sem1_Scores.csv' (Student ID), 'Sem1_Scores.csv' (Full Name), 'Sem1_Scores.csv' (Dept Info), 'Sem1_Scores.csv' (Math), 'Sem1_Scores.csv' (Physics), 'Sem1_Scores.csv' (Chemistry), 'Sem1_Scores.csv' (Programming), and 'Calculation' (PHYSICS_CLEAN). The 'Rows' shelf contains 'Calculation' (PHYSICS_CLEAN).

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The image displays two screenshots of the Tableau interface, showing the 'Sem1_Scores' data source. The top screenshot shows the 'Fields' pane with 'Subject' and 'Score' fields. The bottom screenshot shows the 'Fields' pane with 'Score' field selected. Both screenshots show a data table with columns: Student ID, Full Name, Dept Info, Attendance, Subject, Score, PHYSICS_CLEAN, and Attendance_clean.

Table 1: Sem1_Scores (Top Screenshot)

Student ID	Full Name	Dept Info	Attendance	Subject	Score	PHYSICS_CLEAN	Attendance_clean
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.00
S002	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.00
S003	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.00
S004	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.00
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.00
S002	priya VERMA	ECE-02	90 %	Math	null	null	90.00
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.00
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.00

Table 2: Sem1_Scores (Bottom Screenshot)

Student ID	Full Name	Dept Info	Attendance	Subject	Score	PHYSICS_CLEAN	Attendance_clean
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.00
S002	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.00
S003	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.00
S004	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.00
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.00
S002	priya VERMA	ECE-02	90 %	Math	null	null	90.00
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.00
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.00

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The top screenshot shows the Tableau interface with a calculated field dialog box open. The formula entered is `IFNULL(FLOOR(LENGTH(STR([Score])), ' '), 0)`. The bottom screenshot shows the 'Sem1_Scores' data source connected, with a context menu open over the 'Sem1_Scores.csv' file. The menu options include 'Open...', 'Rename', 'Suggestions...', 'Remove', 'Field names set in first row', 'Generate field names automatically', 'Text File Properties...', and 'Convert to Union...'. The data preview table is also visible.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	PHYSICS_CLEAN
S003	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	0.820000
S004	Sneha Patel	IT-03	92	0.880000	90	96	95 %	0.880000
S205	aditya Rao	MECH-04	45	null	50	60	65 %	0.000000
S006	Aranya Singh	CSE-01	88	0.910000	88	92	88 %	0.910000
S007	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	0.700000
S008	Kavya Reddy	IT-03	96	0.940000	92	98	98 %	0.940000
S009	Vikram Jash	CSE-01	67	0.600000	58	72	80 %	0.600000
S210	Pooja Iyer	ECE-02	82	0.850000	80	89	91 %	0.850000

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The top screenshot shows the 'Union' dialog box in Tableau. The 'Specific (manual)' tab is selected, and two CSV files, 'Sem1_Scores.csv' and 'Sem2_Scores.csv', are listed under 'Connection: Sem1_Scores'. The 'Tables in union: 2' is indicated at the bottom. The 'Apply' button is highlighted.

The bottom screenshot shows the Tableau interface after the union has been created. The 'Connections' pane on the left lists the files. The 'Fields' pane on the right shows the fields from the unioned tables. The 'Columns' shelf contains the following fields: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name, and PPHY.

Type	Field Name	Physical Table	Remote Fl...
Ali	Student_ID	Union	Student_ID
Ali	Full_Name	Union	Full_Name
Ali	Dept_Info	Union	Dept_Info
Ali	Math	Union	Math
Ali	Physics	Union	Physics

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The top screenshot shows the Tableau interface with a calculated field dialog box open. The field is named 'Semester' and contains the following logic:

```
IF CONTAINS([Table Name], "Sem1")
THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2")
THEN "Semester 2" ELSE "Other"
END
```

The bottom screenshot shows the 'Sem1_Scores' data source configuration. The 'Union' relationship is established between 'Sem1_Scores' and 'Department_Master.csv'. The 'Performance Options' are set to 'All'. The resulting data table is as follows:

Student_ID	Full Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name
S001	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	Sem1_Scores.csv
S002	priya VERMA	ECE-02	null	0.750000	80	85	90 %	Sem1_Scores.csv
S003	Rohan Gupta	CSE-01	78	0.820000	null	88	75 %	Sem1_Scores.csv
S004	Sneha Patel	IT-03	92	0.880000	90	96	95 %	Sem1_Scores.csv
S005	aditya Rao	MECH-04	45	null	50	60	65 %	Sem1_Scores.csv
S006	Ananya Singh	CSE-01	88	0.960000	85	92	88 %	Sem1_Scores.csv
S007	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	Sem1_Scores.csv
S008	Kavya Reddy	IT-03	95	0.940000	92	98	96 %	Sem1_Scores.csv

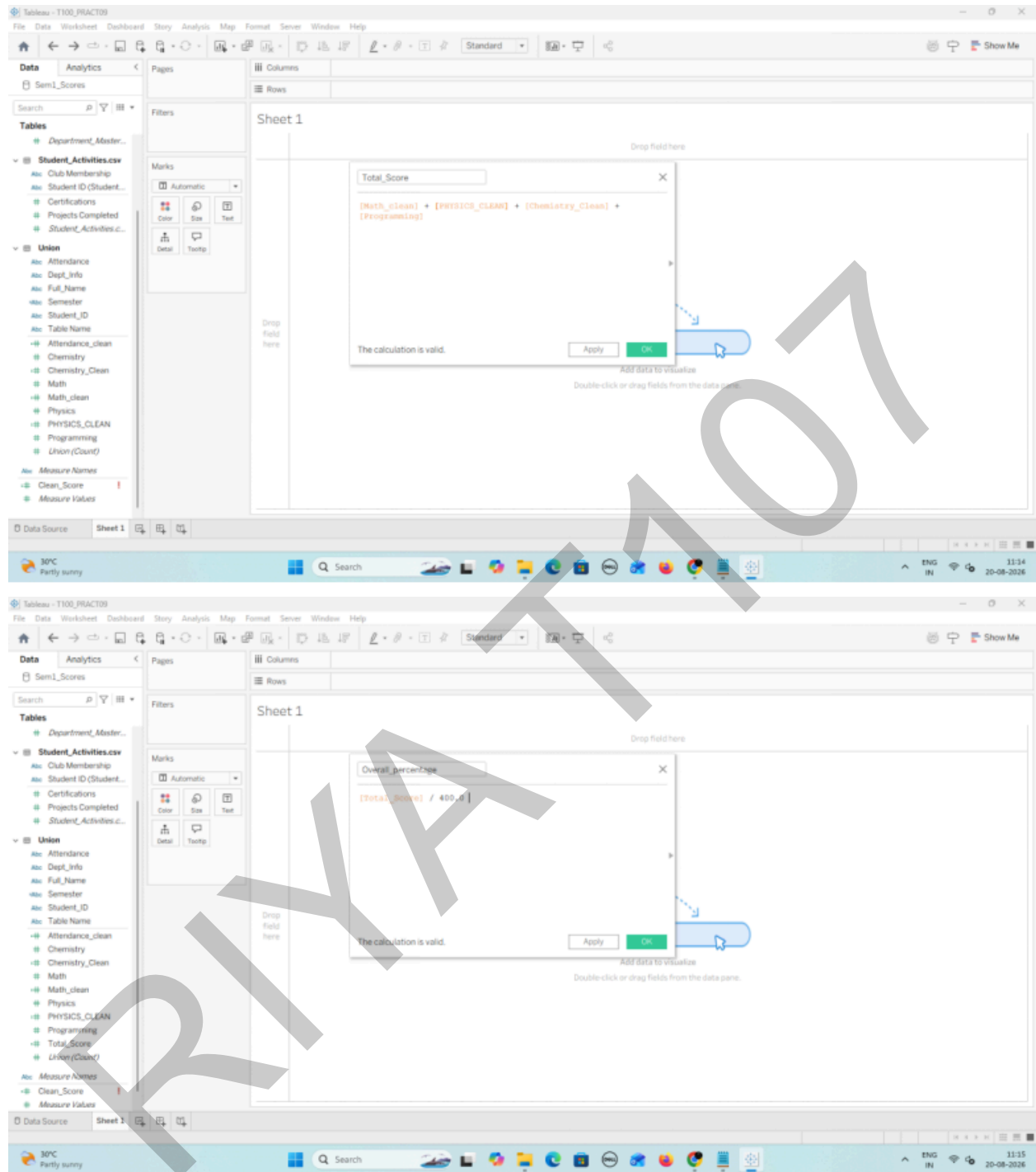
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The top screenshot shows the Tableau interface with the 'Sem1_Scores' data source selected. The 'Connections' pane on the left lists various data sources, including 'Sem1_Scores.csv'. The 'Files' pane on the right shows a list of files, including 'Sem1_Scores.csv'. The 'Data Source' pane on the right shows a union of 'Student_Activities.csv' and 'Student_Activities.csv'. The 'Columns' shelf contains 'Student ID (Student Acti...' and 'Student ID (Stat...'. The 'Rows' shelf contains 'Projects Completed' and 'Certifications'. The 'Marks' shelf is set to 'Automatic'. The 'Performance Options' section is visible at the bottom.

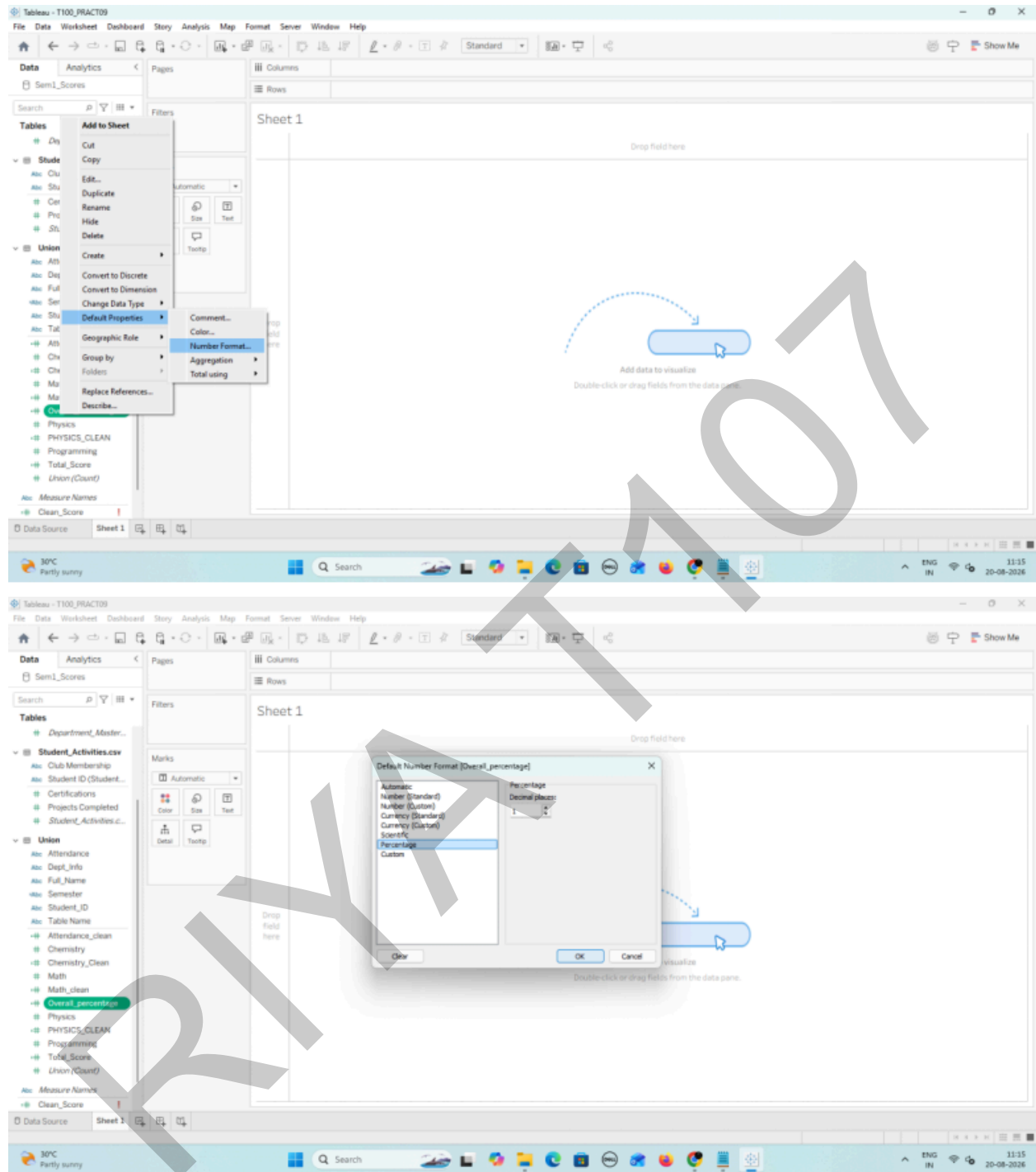
The bottom screenshot shows the Tableau interface with the 'Sem1_Scores' data source selected. The 'Columns' shelf is empty. The 'Rows' shelf is empty. The 'Marks' shelf is set to 'Automatic'. The 'Data Source' pane on the left shows a union of 'Student_Activities.csv' and 'Student_Activities.csv'. The 'Columns' shelf contains 'Student ID (Student Acti...' and 'Student ID (Stat...'. The 'Rows' shelf contains 'Projects Completed' and 'Certifications'. The 'Marks' shelf is set to 'Automatic'. The 'Performance Options' section is visible at the bottom.

- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets

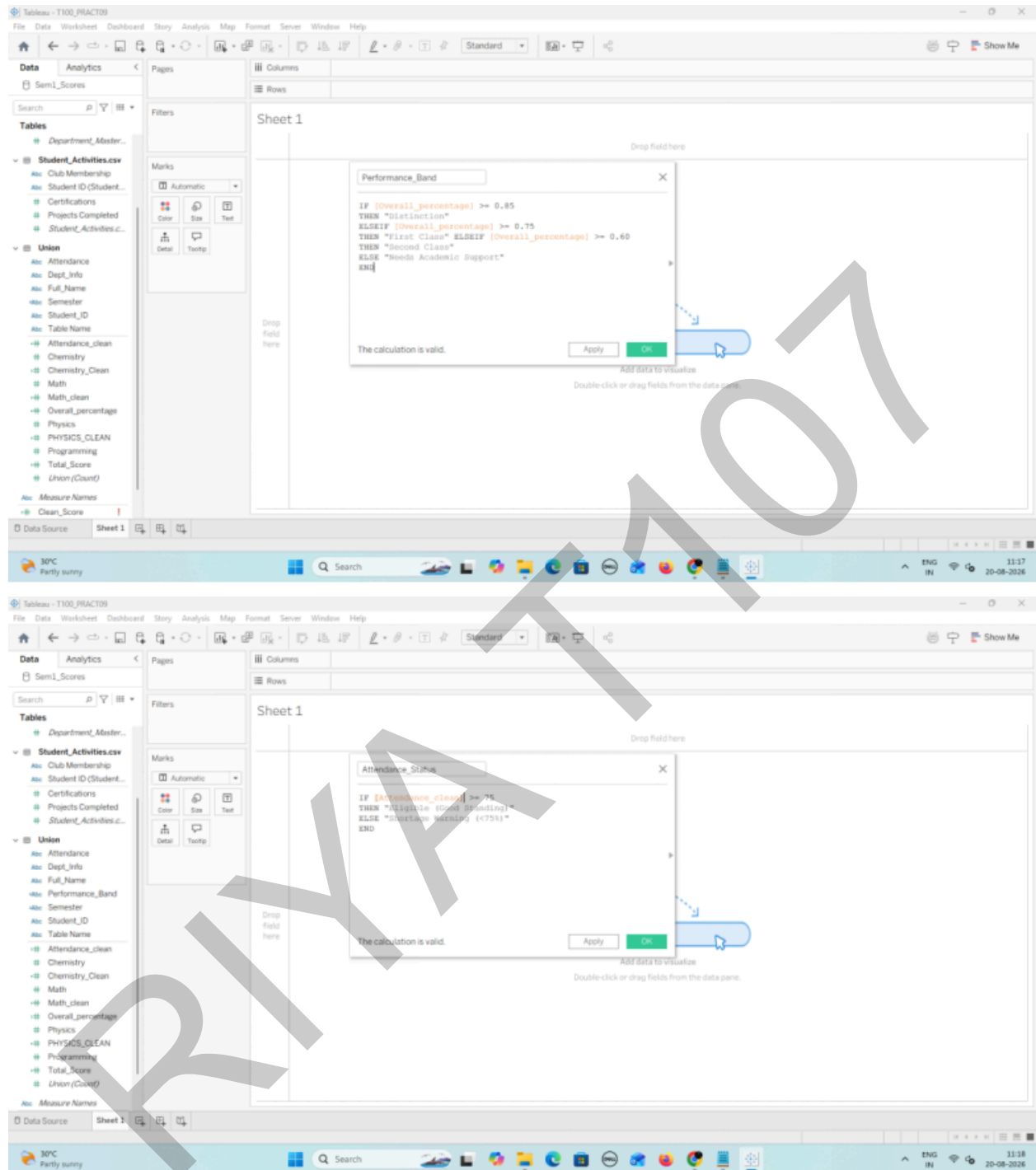
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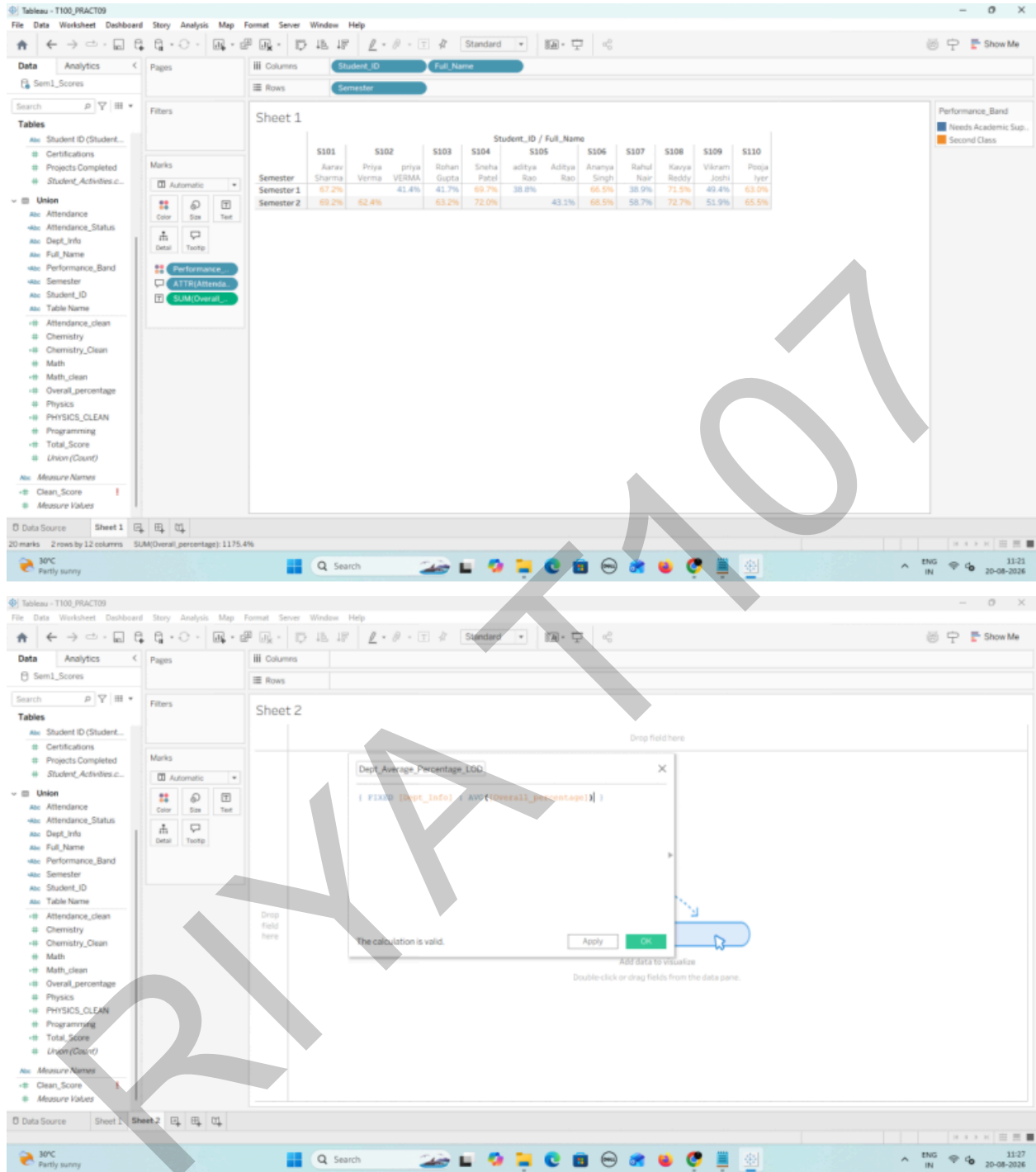
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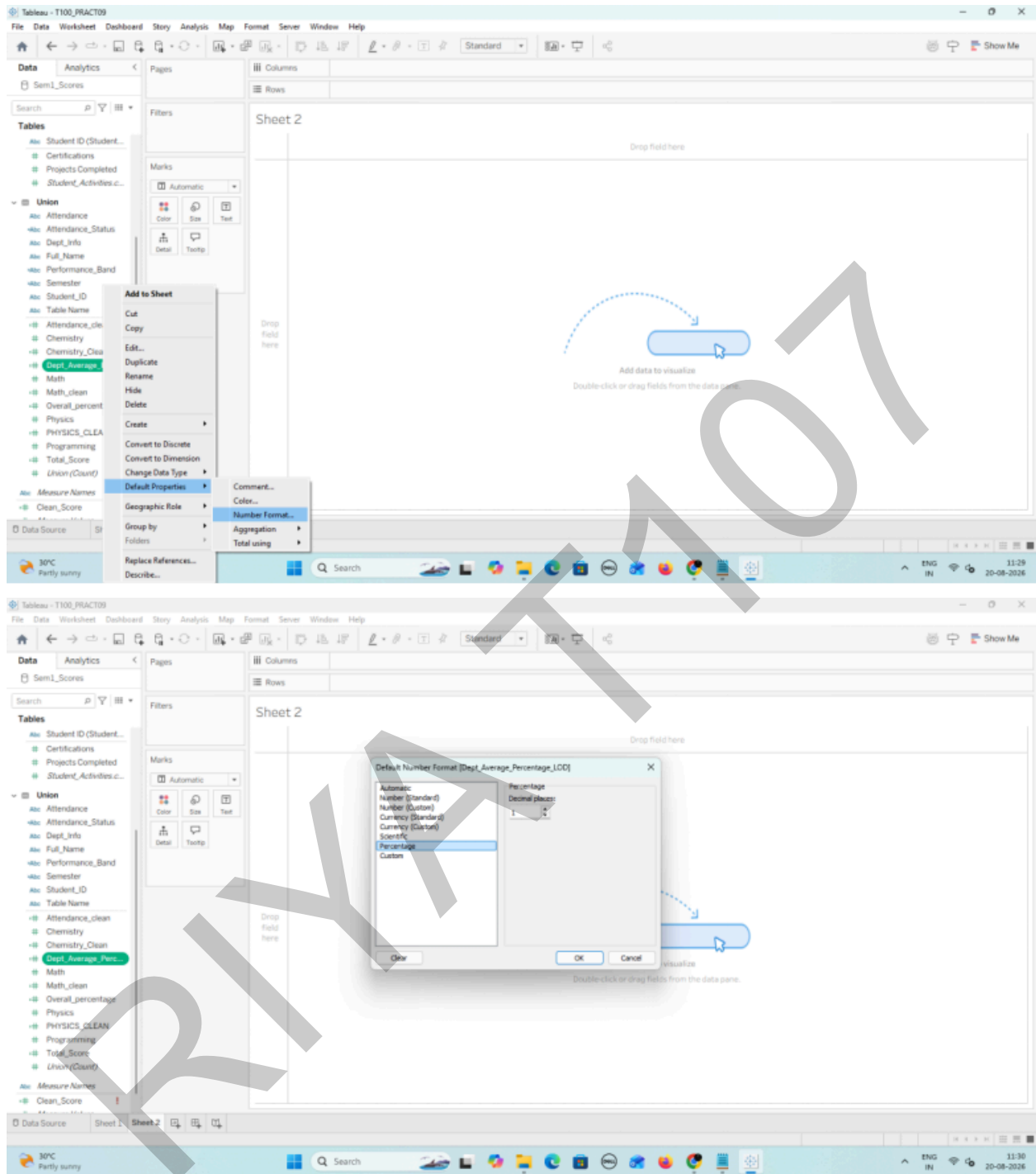
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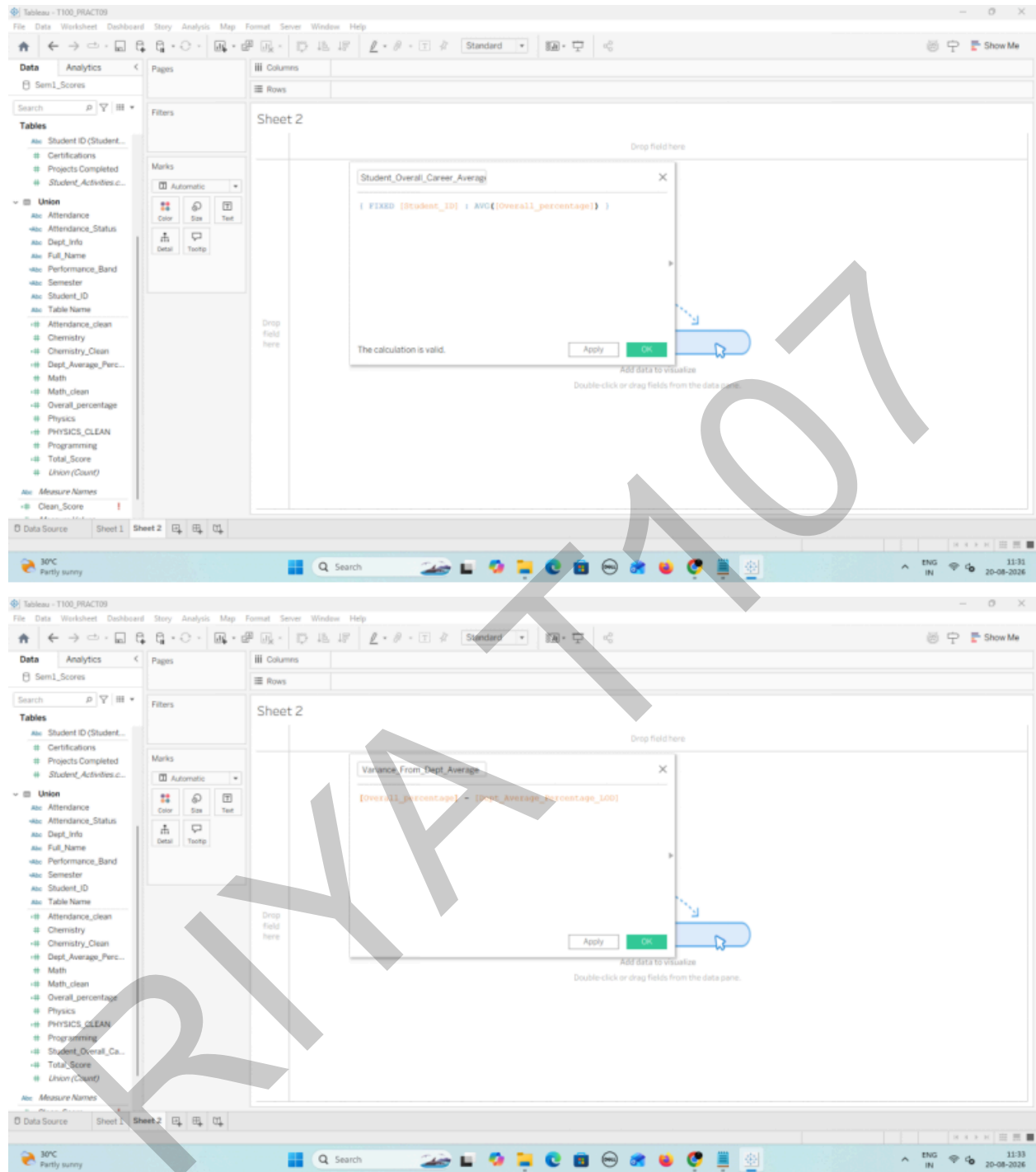
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The first screenshot shows the Tableau interface with a calculated field dialog box open. The dialog contains the following code:

```
IF [Variance_From_Dept_Average] >= 0
THEN "Above Department Benchmark"
ELSE "Below Department Benchmark"
END
```

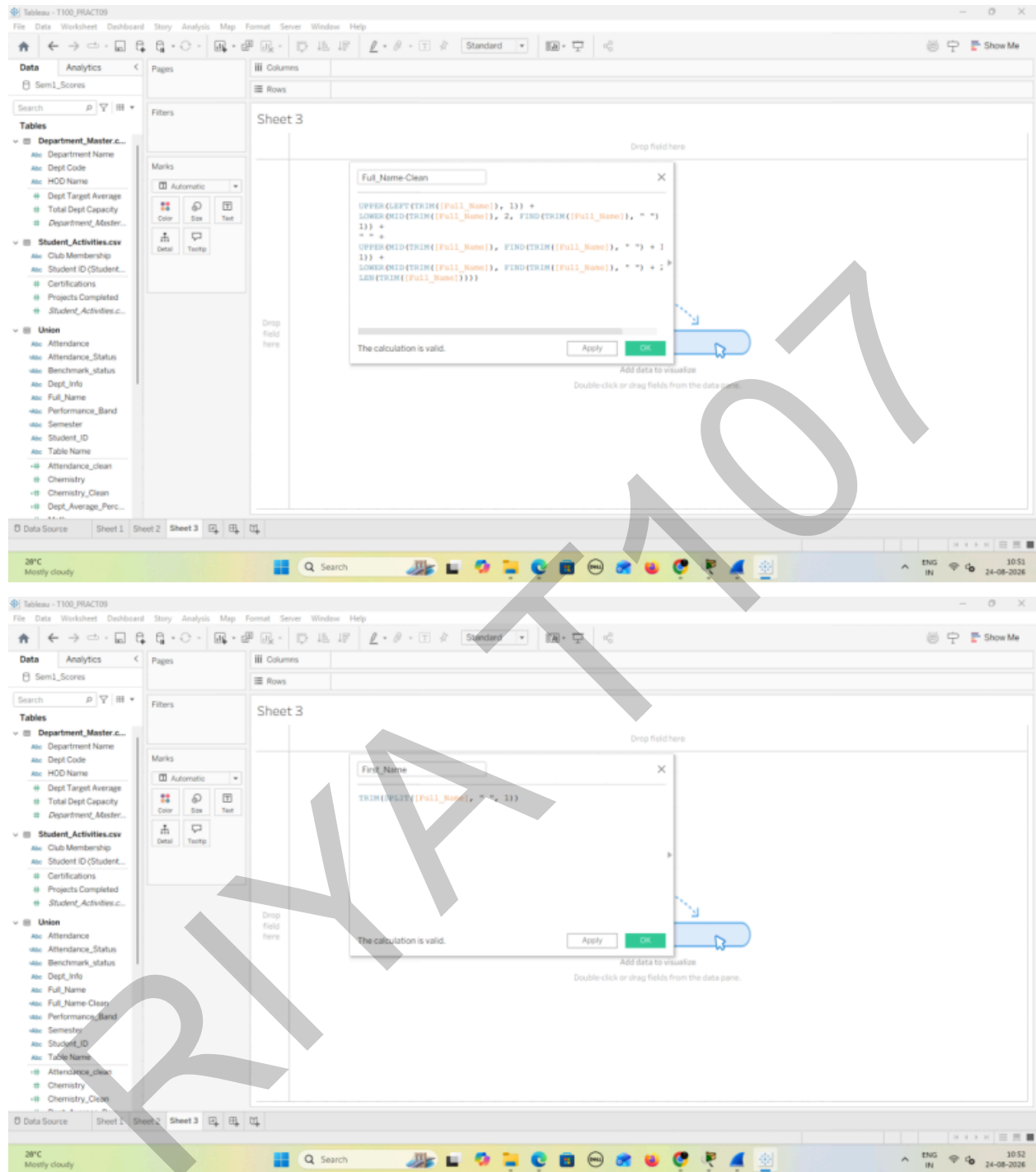
The second screenshot shows the final Tableau view. The calculated field is named "Benchmark_status" and is displayed as a legend on the right side of the view. The view shows a table of student performance data across different departments and semesters.

Dept_Info	Student_ID	Full_Name	Semester	
			r 1	r 2
CSE-01	S101	Aarav Sharma	87.2%	86.2%
	S104	Rohit Gupta	41.2%	63.2%
	S106	Ananya Singh	66.5%	68.6%
	S109	Vikram Joshi	49.4%	51.9%
ECE-02	S102	Priya Verma		62.4%
	S107	Rahul Nair	56.9%	58.7%
	S110	Pooja Iyer	63.0%	65.9%
IT-03	S108	Snigdha Patel	68.7%	72.0%
	S105	Kavya Reddy	75.5%	72.7%
MECH-04	S103	Aditya Rao	57.8%	49.1%
	S105	Aditya Rao		

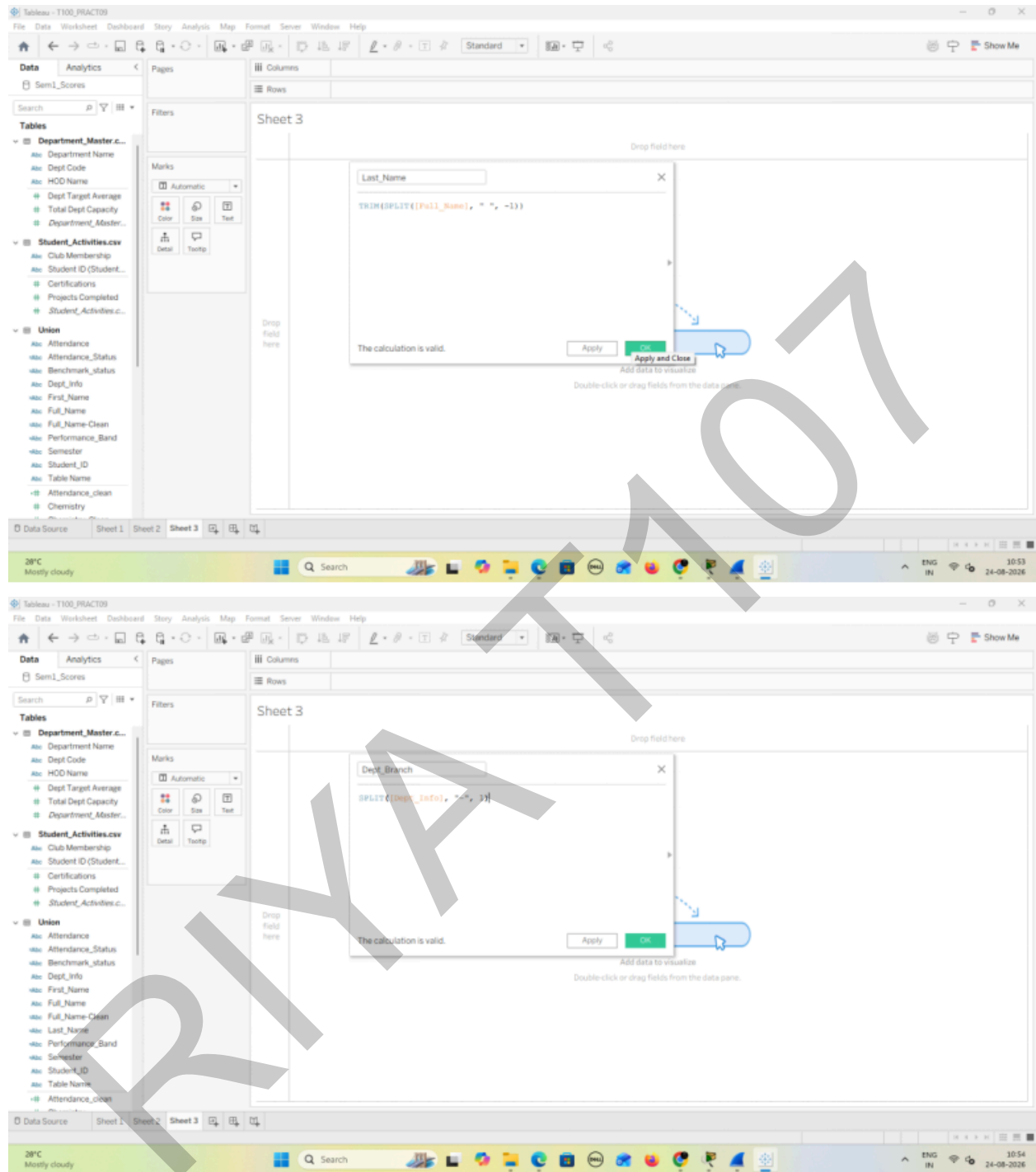
g) Apply advanced data cleaning techniques

RIYA RANE T107
PRACTICAL NO : 9

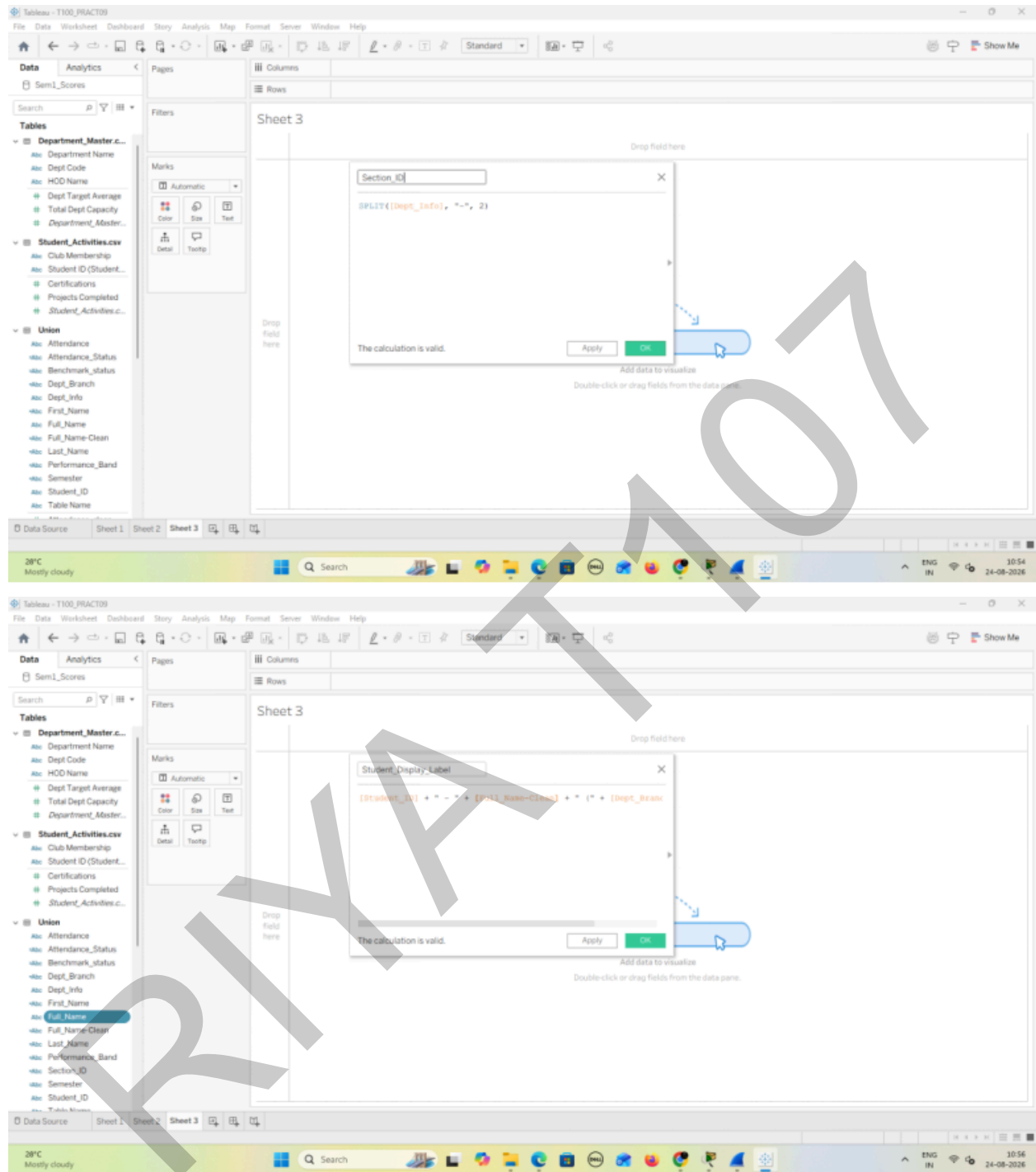
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